



2014 Assessment Examination

FORM VI

MATHEMATICS EXTENSION 1

Monday 19th May 2014

General Instructions

- Writing time — 1 hour 30 minutes
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 55 Marks

- All questions may be attempted.

Section I – 7 Marks

- Questions 1–7 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 48 Marks

- Questions 8–11 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eight.
- Write your candidate number on this question paper and submit it with your answers.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Candidature — 123 boys

Examiner
LRP

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

The exact value of $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)\cos^{-1}\left(\frac{1}{2}\right)$ in radians is:

1

- (A) 0.822
- (B) 2700
- (C) $\frac{\pi^2}{12}$
- (D) $\frac{\pi^2}{24}$

QUESTION TWO

If $x + \alpha$ is a factor of $7x^3 + 9x^2 - 5\alpha x$, where $\alpha \neq 0$, then the value of α is:

1

- (A) 2
- (B) $\frac{4}{7}$
- (C) $-\frac{4}{7}$
- (D) -2

QUESTION THREE

A particle is moving in a straight line in such a way that its displacement, x metres, at time t seconds is given by $x = 2.5t + 5 \cos(0.5t)$, where $t \geq 0$.

1

The minimum velocity of the particle, in metres per second, is:

- (A) -5
- (B) -2.5
- (C) 0
- (D) 2.5

QUESTION FOUR

If $f'(x) = \cos x$ and $f\left(\frac{\pi}{2}\right) = 0$, then $f(x)$ is equal to:

1

- (A) $\sin x$
- (B) $1 - \sin x$
- (C) $\sin x - 1$
- (D) $\sin x + 1$

QUESTION FIVE

Which one of the following is **not** equal to $\tan\left(\frac{\pi}{5}\right)$?

1

- (A) $\frac{1}{\cot\left(\frac{\pi}{5}\right)}$
- (B) $\cot\left(\frac{3\pi}{10}\right)$
- (C) $\frac{2 \tan\left(\frac{\pi}{10}\right)}{1 - \tan^2\left(\frac{\pi}{10}\right)}$
- (D) $\frac{2 \tan\left(\frac{2\pi}{5}\right)}{1 - \tan^2\left(\frac{2\pi}{5}\right)}$

QUESTION SIX

The velocity v of a particle as a function of its displacement x is given by $v = \frac{2}{\sqrt{1-x^2}}$.

1

Its acceleration is:

- (A) $2 \sin^{-1} x$
- (B) $\frac{4x}{(1-x^2)^2}$
- (C) $\frac{2x}{(1-x^2)^2}$
- (D) $\frac{2x}{(1-x^2)^{\frac{3}{2}}}$

QUESTION SEVEN

The graphs of $y = ax$ and $y = \tan^{-1}(bx)$ intersect exactly **three** times if:

1

- (A) $a = b$
- (B) $a = -b$
- (C) $0 < b < a$
- (D) $0 < a < b$

————— End of Section I —————

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION EIGHT (12 marks) Use a separate writing booklet.

Marks

(a) Let $f(x) = 2 \cos^{-1} \left(\frac{x}{3} \right)$. What is the domain of $f(x)$? 1

(b) Find the exact value of $\int_0^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$. 2

(c) Use the expansion of $\cos(A - B)$ to show that $\cos 15^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}}$. 2

(d) Solve the equation $2 \sin^2 \theta = \sin 2\theta$ for $0 \leq \theta \leq 2\pi$. 3

(e) A particle is moving in simple harmonic motion according to the equation

$$x = 2 \cos \left(\frac{\pi}{10} t \right) + 3,$$

where x is the displacement in metres at time t seconds.

(i) Find the amplitude and period of the motion. 2

(ii) Sketch the displacement-time graph over the first 60 seconds. 2

QUESTION NINE (12 marks) Use a separate writing booklet.

Marks

- (a) The acceleration of a particle moving in a straight line is given by $\ddot{x} = 10x - 4x^3$, where x is the displacement in metres at time t seconds. Initially the particle is stationary at $x = \sqrt{6}$ metres.

(i) Find the initial acceleration.

1

(ii) Find an expression for v^2 as a function of x .

2

- (b) Consider the polynomial $P(x) = x^4 - x^3 - 3x^2 + 5x - 2$.

(i) Show that 1 and -2 are zeroes of $P(x)$.

1

(ii) Factorise $P(x)$ into linear factors.

3

(iii) Without the aid of calculus, sketch the graph of $P(x)$, clearly indicating all intercepts with the axes.

2

- (c) (i) Differentiate $x \sin^{-1} x + \sqrt{1 - x^2}$.

2

(ii) Hence evaluate $\int_0^1 \sin^{-1} x \, dx$.

1

QUESTION TEN (12 marks) Use a separate writing booklet.

Marks

- (a) Using the fact that $\cos 3x = 4 \cos^3 x - 3 \cos x$, find general solutions of the equation $\cos 3x + 2 \cos x = 0$. 4
- (b) A particle is moving in simple harmonic motion according to $x = 2 \sin 3t - 2\sqrt{3} \cos 3t$, where x is its displacement in metres from a fixed point O at time t seconds.
- (i) Express x in the form $x = R \sin(3t - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. 2
- (ii) Find the initial displacement and velocity. 2
- (iii) Find the first time that the particle is 2 metres from O . 2
- (c) A body is moving with velocity $v = 4 - 2t \text{ ms}^{-1}$, where t is time in seconds. Find the total distance it travels in the first 5 seconds of its motion. 2

QUESTION ELEVEN (12 marks) Use a separate writing booklet.

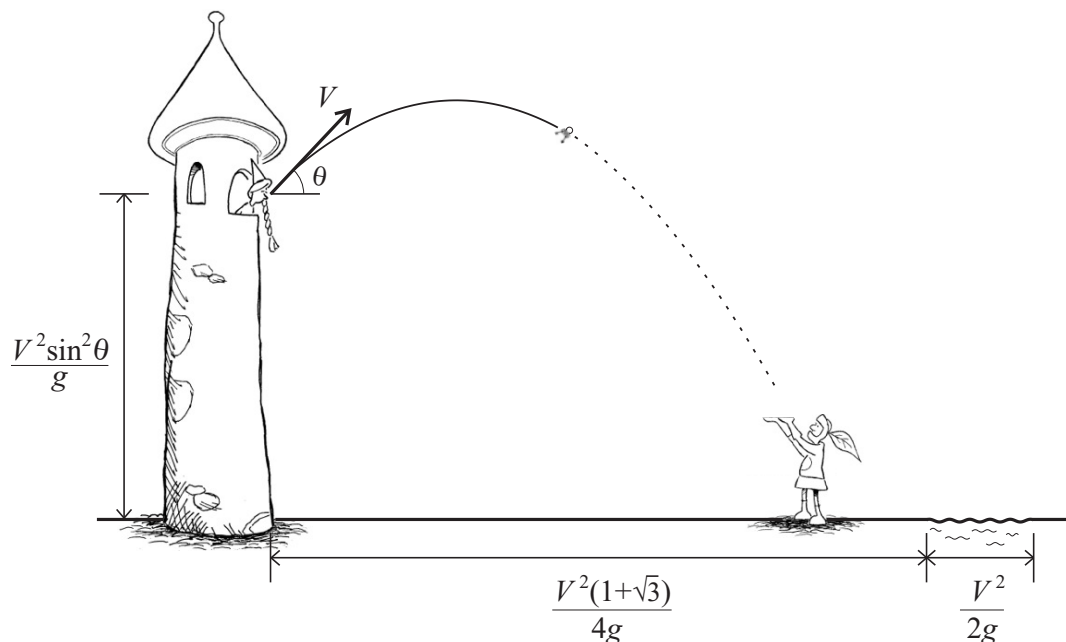
Marks

- (a) If $\tan \alpha$ and $\tan \beta$ are two values of $\tan \theta$ which satisfy the quadratic equation $a \tan^2 \theta + b \tan \theta + c = 0$:
- (i) Find $\tan(\alpha + \beta)$ in terms of a , b and c . 2
- (ii) Show that $\tan^2(\alpha - \beta) = \frac{b^2 - 4ac}{(a + c)^2}$. 2

QUESTION ELEVEN CONTINUES ON THE NEXT PAGE

QUESTION ELEVEN (Continued)

(b)



Rapunzel is trapped on the top floor of an enchanted tower. She throws a set of keys that unlock the tower to a handsome prince standing on the ground below. The keys are projected with an initial velocity $V \text{ ms}^{-1}$ and at an angle θ to the horizontal, where $0^\circ < \theta < 90^\circ$.

The point of projection is a window located $\frac{V^2 \sin^2 \theta}{g}$ metres above the ground, where g is the acceleration due to gravity. Assume that the prince can't catch and the keys fall to the ground. The horizontal and vertical displacement equations of the keys at time t are given by

$$x = Vt \cos \theta \quad \text{and} \quad y = Vt \sin \theta - \frac{gt^2}{2} + \frac{V^2 \sin^2 \theta}{g}.$$

(i) Show that the Cartesian equation of the path of the keys is given by

1

$$y = x \tan \theta - \frac{gx^2 \sec^2 \theta}{2V^2} + \frac{V^2 \sin^2 \theta}{g}.$$

(ii) Show that the horizontal range of the keys is given by $\frac{V^2 (1 + \sqrt{3}) \sin 2\theta}{2g}$ metres.

3

(iii) The near edge of a moat lies $\frac{V^2 (1 + \sqrt{3})}{4g}$ metres from the base of the tower.

4

The moat is $\frac{V^2}{2g}$ metres wide. Find the values of θ for which the keys will land either side of the moat.

End of Section II

END OF EXAMINATION

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



2014
Assessment Examination
FORM VI
MATHEMATICS EXTENSION 1
Monday 19th May 2014

- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

CANDIDATE NUMBER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

2014 Form VI May Assessment

SOLUTIONS

* MULTIPLE CHOICE

Q1 C

Q2 A

Q3 C

Q4 C

Q5 D

Q6 B

Q7 D

Multiple Choice Working:

Q1. $\frac{\pi}{4} \times \frac{\pi}{3} = \frac{\pi^2}{12}$ C

Q2. $7(-\alpha)^3 + 9(-\alpha)^2 - 5\alpha(-\alpha) = 0$
 $-7\alpha^3 + 9\alpha^2 + 5\alpha^2 = 0$
 $-\alpha^2(7\alpha - 14) = 0$
 $\therefore \alpha = 2 \quad [\alpha \neq 0]$ A

Q3. $\dot{x} = 2.5 - 2.5 \sin(0.5t)$
 $\therefore \dot{x}_{\min} = 0$ C

Q4. $f(x) = \sin x + C$
 $0 = 1 + C \Rightarrow C = -1$
 $\therefore f(x) = \sin x - 1$ C

Q5. A: $\cot \frac{\pi}{5} = \tan \frac{\pi}{5}$
B: $\cot(\frac{3\pi}{10}) = \tan(\frac{\pi}{2} - \frac{3\pi}{10}) = \tan(\frac{\pi}{5})$
C: $\frac{2 \tan(\frac{\pi}{10})}{1 - \tan^2(\frac{\pi}{10})} = \tan(2 \times \frac{\pi}{10}) = \tan(\frac{\pi}{5})$ D

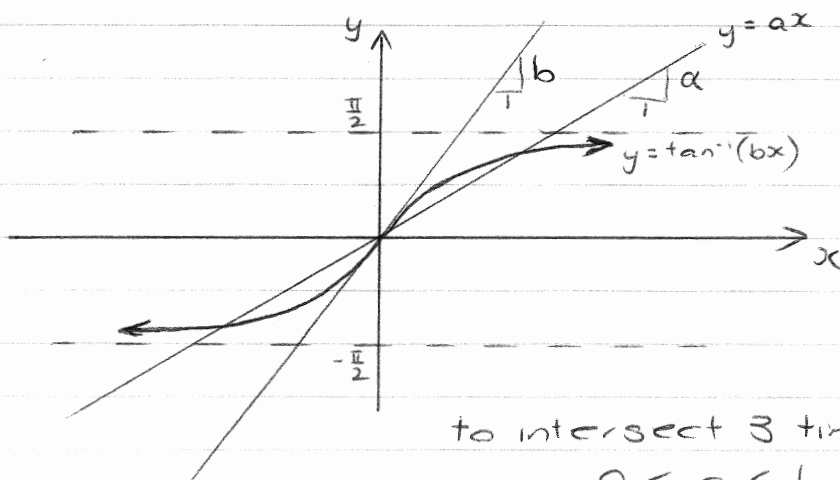
$\rightarrow$ could have also just put it into a calculator...

$$\begin{aligned}
 \text{Q6. } \dot{x} &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\
 &= \frac{d}{dx} \left(\frac{1}{2} \cdot \frac{4}{1-x^2} \right) \\
 &= \frac{d}{dx} \left(2(1-x^2)^{-1} \right) \\
 &= -2(1-x^2)^{-2} x - 2x \\
 &= \frac{4x}{(1-x^2)^2}
 \end{aligned}$$

B

$$\text{Q7. } y = ax \rightarrow \frac{dy}{dx} = a \rightarrow m = a$$

$$y = \tan^{-1}(bx) \rightarrow \begin{cases} \frac{dy}{dx} = \frac{b}{1+b^2x^2} \\ \text{when } x=0, m_T = b \end{cases}$$



to intersect 3 times D
 $0 < a < b$
 OR $b < a < 0$

QUESTION 8:

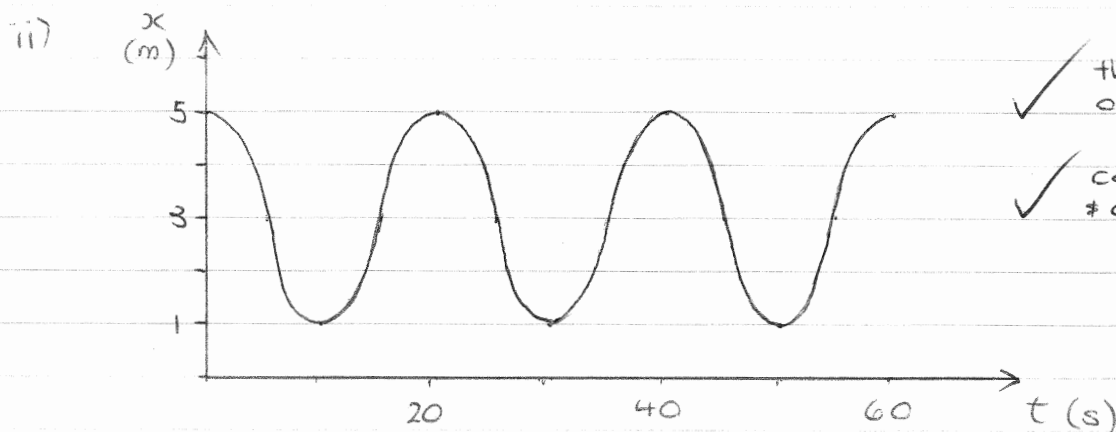
a) $-1 \leq \frac{x}{3} \leq 1 \quad \therefore \text{Domain: } -3 \leq x \leq 3 \quad \checkmark$

b) $\int_0^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx = \left[\sin^{-1} \frac{x}{2} \right]_0^{\sqrt{3}} \quad \checkmark$
 $= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} 0$
 $= \frac{\pi}{3} \quad \checkmark$

c) $\cos 15^\circ = \cos(45^\circ - 30^\circ)$
 $= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \quad \checkmark$
 $= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2} \quad \checkmark$
 $= \frac{\sqrt{3} + 1}{2\sqrt{2}} \quad \text{as required} \quad \checkmark$

d) $2 \sin^2 \theta = \sin 2\theta$
 $2 \sin \theta (\sin \theta - \cos \theta) = 0 \quad \checkmark$
 $\therefore 2 \sin \theta = 0 \quad \text{or} \quad \sin \theta - \cos \theta = 0$
 $\sin \theta = 0 \quad \tan \theta = 1 \quad [\cos \theta \neq 0]$
 $\therefore \theta = 0, \pi, 2\pi \quad \theta = \frac{\pi}{4}, \frac{5\pi}{4} \quad \checkmark$
 $\therefore \theta = 0, \frac{\pi}{4}, \pi, \frac{5\pi}{4} \text{ or } 2\pi$

e) i) Amplitude = 2 metres $\checkmark$
 Period = $2\pi \div \frac{\pi}{10}$
 $= 20 \text{ seconds} \quad \checkmark$



QUESTION 9:

a) When $t = 0$: $v = 0 \text{ ms}^{-1}$
 $x = \sqrt{6} \text{ m}$

i) $\ddot{x} = 10x - 4x^3$
 $= 10(\sqrt{6}) - 4(\sqrt{6})^3$
 $= 10\sqrt{6} - 24\sqrt{6}$
 $= -14\sqrt{6} \text{ ms}^{-2} \quad \checkmark \quad (\text{exact value})$

ii) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 10x - 4x^3$

$\frac{1}{2} v^2 = 5x^2 - x^4 + C \quad \checkmark$

at $x = \sqrt{6}$, $v = 0$:

$0 = 5(\sqrt{6})^2 - (\sqrt{6})^4 + C$

$\therefore C = 6$

$\therefore v^2 = 10x^2 - 2x^4 + 12 \quad \checkmark$

b) i) $P(1) = (1)^4 - (1)^3 - 3(1)^2 + 5(1) - 2$
 $= 0$

$P(-2) = (-2)^4 - (-2)^3 - 3(-2)^2 + 5(-2) - 2$
 $= 0$

} $\checkmark$

ii) METHOD 1:

$x^4 - x^3 - 3x^2 + 5x - 2 = (x-1)(x+2)(x^2 + kx + 1)$
 $= (x^2 + x - 2)(x^2 + kx + 1) \quad \checkmark$

Equating coeff's of x^3 :

$-1 = k + 1$

$\therefore k = -2 \quad \checkmark$

$\therefore P(x) = (x-1)(x+2)(x^2 - 2x + 1)$
 $= (x-1)^3(x+2) \quad \checkmark$

OR

METHOD 2:

Let the zeroes be 1, -2, $\alpha \neq \beta$

$$\text{Sum of roots: } 1 + (-2) + \alpha + \beta = -\frac{(-1)}{1}$$

$$\alpha + \beta = 2 \quad \text{--- ①}$$

$$\text{Product of roots: } 1 \times (-2) \times \alpha \times \beta = \frac{-2}{1}$$

$$\alpha\beta = 1 \quad \text{--- ②}$$

✓ either

$$\text{Rearrange ①: } \beta = 2 - \alpha \quad \text{①*}$$

$$\text{Sub ①* into ②: } \alpha(2 - \alpha) = 1$$

$$\alpha^2 - 2\alpha + 1 = 0$$

$$(\alpha - 1)^2 = 0$$

$$\therefore \alpha = 1 \quad \checkmark$$

$$\text{Sub into ①* : } \beta = 2 - 1$$

$$= 1 \quad \checkmark$$

$$\therefore P(x) = (x - 1)^3(x + 2) \quad \checkmark$$

OR

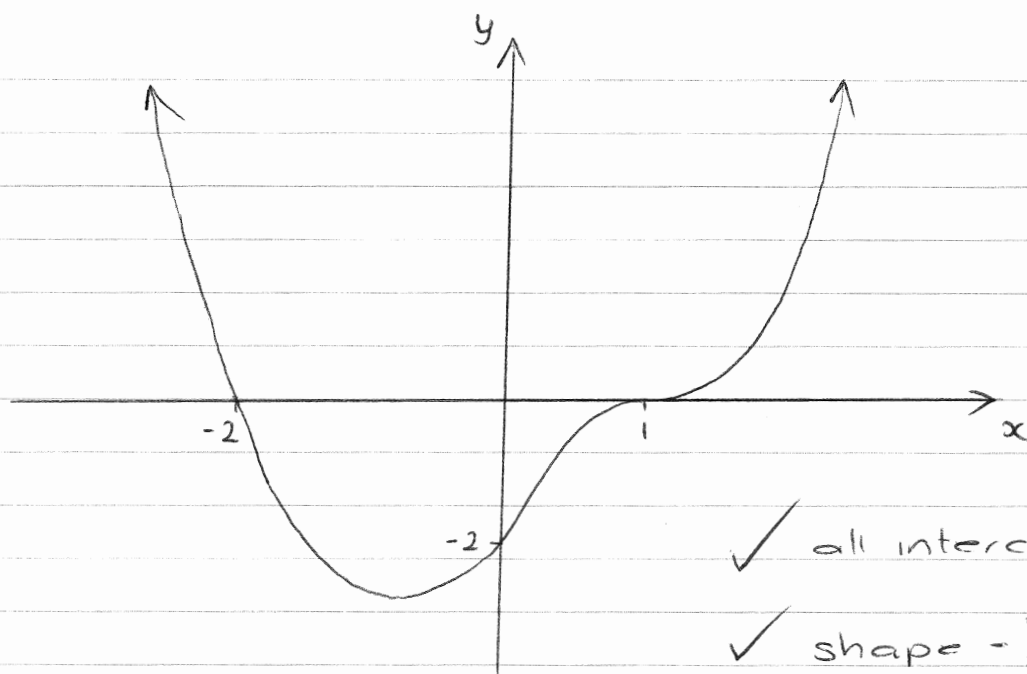
METHOD 3:

$$(x - 1)(x + 2) = x^2 + x - 2$$

$$\begin{array}{r}
 x^2 + x - 2 \quad \checkmark \quad \left\{ \begin{array}{l}
 x^2 - 2x + 1 \\
 \hline
 x^4 - x^3 - 3x^2 + 5x - 2 \\
 x^4 + x^3 - 2x^2 \\
 \hline
 -2x^3 - x^2 + 5x \\
 -2x^3 - 2x^2 + 4x \\
 \hline
 x^2 + x - 2 \\
 x^2 + x - 2 \\
 \hline
 0
 \end{array} \right.
 \end{array}
 \quad \checkmark$$

$$\begin{aligned}
 \therefore P(x) &= (x - 1)(x + 2)(x^2 - 2x + 1) \\
 &= (x - 1)^3(x + 2) \quad \checkmark
 \end{aligned}$$

(iii)



✓ all intercepts

✓ shape - horizontal point of inflexion at 1 must be clear

$$c) i) \frac{d}{dx} (x \sin^{-1} x + \sqrt{1-x^2})$$

$$= \sin^{-1} x \times 1 + x \times \frac{1}{\sqrt{1-x^2}} + \frac{1}{2} (1-x^2)^{-\frac{1}{2}} \times -2x$$

$$= \sin^{-1} x + \frac{x}{\sqrt{1-x^2}} - \frac{x}{\sqrt{1-x^2}} \quad \text{(for either)}$$

$$= \sin^{-1} x$$

$$ii) \int_0^1 \sin^{-1} x \, dx = [x \sin^{-1} x + \sqrt{1-x^2}]_0^1$$

$$= 1 \times \frac{\pi}{2} + 0 - (0 + 1)$$

$$= \frac{\pi}{2} - 1$$

✓

QUESTION 10:

a) $4 \cos^3 x - 3 \cos x + 2 \cos x = 0$

$$4 \cos^3 x - \cos x = 0$$

$$\cos x (4 \cos^2 x - 1) = 0 \quad \checkmark$$

$$\cos x (2 \cos x - 1)(2 \cos x + 1) = 0$$

$$\therefore \cos x = 0, \frac{1}{2} \text{ or } -\frac{1}{2} \quad \checkmark$$

General solutions:

$$\cos x = 0 \rightarrow x = 2n\pi \pm \frac{\pi}{2} \quad (n \in \mathbb{Z}) \quad \checkmark$$

$$\cos x = \frac{1}{2} \rightarrow x = 2n\pi \pm \frac{\pi}{3} \quad (n \in \mathbb{Z})$$

$$\cos x = -\frac{1}{2} \rightarrow x = 2n\pi \pm \frac{2\pi}{3} \quad (n \in \mathbb{Z})$$

Alternatively:

$$\cos x = 0 \rightarrow x = \frac{\pi}{2} + n\pi$$

$$\cos x = \pm \frac{1}{2} \rightarrow x = n\pi \pm \frac{\pi}{3}$$

b) i) $2 \sin 3t - 2\sqrt{3} \cos 3t = R \sin(3t - \alpha)$

$$= R \sin 3t \cos \alpha - R \cos 3t \sin \alpha$$

$$R \cos \alpha = 2$$

$$R \sin \alpha = -2\sqrt{3}$$

} $\checkmark$

$$R^2 = 2^2 + (2\sqrt{3})^2$$

$$= 16$$

$$\therefore R = 4$$

$$\tan \alpha = \frac{2\sqrt{3}}{2}$$

$$= \sqrt{3}$$

$$\therefore \alpha = \frac{\pi}{3}$$

$$\therefore x = 4 \sin\left(3t - \frac{\pi}{3}\right) \quad \checkmark$$

ii) $\dot{x} = 12 \cos\left(3t - \frac{\pi}{3}\right)$

when $t=0$: $x = 4 \sin\left(-\frac{\pi}{3}\right)$

$$= -2\sqrt{3} \text{ m} \quad \checkmark$$

$$\dot{x} = 12 \cos\left(-\frac{\pi}{3}\right)$$

$$= 6 \text{ m/s} \quad \checkmark$$

iii) since starts at $x = -2\sqrt{3}$ m ...

first time 2m from origin $\rightarrow x = -2$ m

$$-2 = 4 \sin\left(3t - \frac{\pi}{3}\right)$$

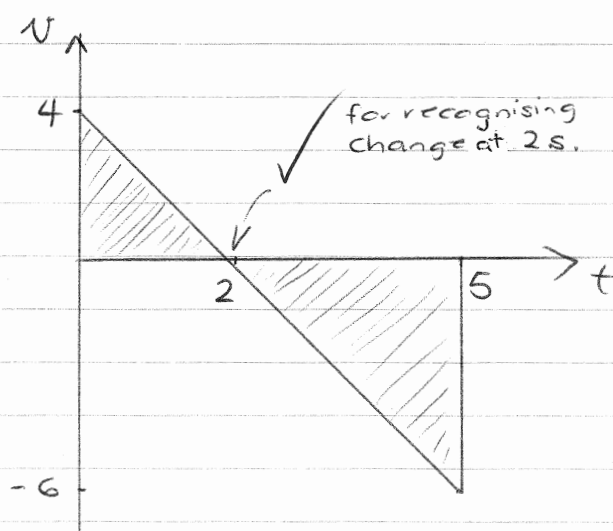
$$\sin\left(3t - \frac{\pi}{3}\right) = -\frac{1}{2}$$

$$3t - \frac{\pi}{3} = -\frac{\pi}{6}, \frac{7\pi}{6}, \dots$$

$$t = \frac{\pi}{18}, \frac{\pi}{2}, \dots$$

$\therefore$ the first time is at $t = \frac{\pi}{18}$ s. ✓

c)



distance = total area travelled

$$= \frac{1}{2} \times 2 \times 4 + \frac{1}{2} \times 3 \times 6$$

$$= 4 + 9$$

$$= 13 \text{ m} \quad \checkmark$$

Alternatively:

$$\dot{x} = 4 - 2t$$

$$x = 4t - t^2 + C$$

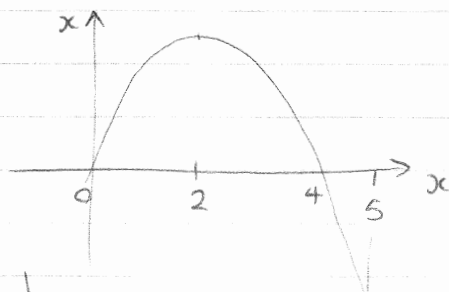
$$= t(4 - t) + C$$

$$\text{Total distance} = 2 \times x_2 + |x_5|$$

$$= 2(8 - 4) + |20 - 25|$$

$$= 8 + 5$$

$$= 13 \text{ m.}$$



QUESTION 11:

$$a) i) \tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$= \frac{-\frac{b}{a}}{1 - \frac{c}{a}}$$

$$= \frac{b}{c - a}$$

$$ii) \tan^2(\alpha - \beta) = \left(\frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \right)^2$$

$$= \frac{\tan^2 \alpha - 2 \tan \alpha \tan \beta + \tan^2 \beta}{(1 + \tan \alpha \tan \beta)^2}$$

$$= \frac{(\tan \alpha + \tan \beta)^2 - 4 \tan \alpha \tan \beta}{(1 + \tan \alpha \tan \beta)^2}$$

$$= \frac{\left(-\frac{b}{a}\right)^2 - 4\left(\frac{c}{a}\right)}{\left(1 + \frac{c}{a}\right)^2}$$

$$= \frac{\frac{b^2}{a^2} - \frac{4c}{a}}{\left(1 + \frac{c}{a}\right)^2} \times \frac{a^2}{a^2}$$

$$= \frac{b^2 - 4ac}{(a + c)^2}$$

$$b) i) t = \frac{x}{v \cos \theta}$$

$$y = v \times \frac{x}{v \cos \theta} \times \sin \theta - \frac{g}{2} \times \frac{x^2}{v^2 \cos^2 \theta} + \frac{v^2 \sin^2 \theta}{g} \quad \checkmark$$

$$= x \tan \theta - \frac{g x^2 \sec^2 \theta}{2 v^2} + \frac{v^2 \sin^2 \theta}{g} \quad \text{as required}$$

ii) The keys will hit the ground when $y=0$:

METHOD 1:

$$0 = vt \sin \theta - \frac{gt^2}{2} + \frac{v^2 \sin^2 \theta}{g} \quad \checkmark$$

$$a = \frac{g}{2}$$

$$b = v \sin \theta$$

$$c = \frac{v^2 \sin^2 \theta}{g}$$

$$t = \frac{-v \sin \theta \pm \sqrt{v^2 \sin^2 \theta - 4 \times -\frac{g}{2} \times \frac{v^2 \sin^2 \theta}{g}}}{2 \times -\frac{g}{2}} \quad \checkmark$$

$$= \frac{-v \sin \theta \pm v \sin \theta \sqrt{3}}{-g}$$

$$= \frac{v \sin \theta (1 + \sqrt{3})}{g} \quad (\text{since } t > 0)$$

$$x = \frac{v \cos \theta \times v \sin \theta (1 + \sqrt{3})}{g}$$

$$= \frac{v^2 (1 + \sqrt{3}) \sin 2\theta}{2g} \quad \text{as required.} \quad \checkmark$$

METHOD 2:

$$0 = x \tan \theta - \frac{gx^2 \sec^2 \theta}{2v^2} + \frac{v^2 \sin^2 \theta}{g}$$

$$a = -\frac{g \sec^2 \theta}{2v^2}$$

$$b = \tan \theta$$

$$c = \frac{v^2 \sin^2 \theta}{g}$$

$$x = \frac{-\tan \theta \pm \sqrt{\tan^2 \theta - 4 \times -\frac{g \sec^2 \theta}{2v^2} \times \frac{v^2 \sin^2 \theta}{g}}}{2 \times -\frac{g \sec^2 \theta}{2v^2}}$$

$$= \frac{(-\tan \theta \pm \tan \theta \sqrt{3}) \times v^2}{-g \sec^2 \theta}$$

$$= \frac{v^2 \tan \theta (1 + \sqrt{3}) \cos^2 \theta}{g} \quad (\text{since } x > 0)$$

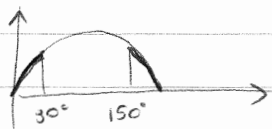
$$= \frac{v^2 \sin \theta \cos \theta (1 + \sqrt{3})}{g} \quad \left(\text{since } \tan \theta = \frac{\sin \theta}{\cos \theta} \right)$$

$$= \frac{v^2 (1 + \sqrt{3}) \sin 2\theta}{2g} \quad \text{as required} \quad \left(\text{since } \sin \theta \cos \theta = \frac{\sin 2\theta}{2} \right)$$

iii) * Near side of moat:

$$0 < \frac{v^2(1+\sqrt{3})\sin 2\theta}{2g} < \frac{v^2(1+\sqrt{3})}{4g} \quad \checkmark$$

$$0 < \sin 2\theta < \frac{1}{2} \quad \text{for } 0^\circ < 2\theta < 180^\circ$$



$$0^\circ < 2\theta < 30^\circ \quad \text{or} \quad 150^\circ < 2\theta < 180^\circ$$

$$0 < \theta < 15^\circ \quad \quad \quad 75^\circ < \theta < 90^\circ \quad \checkmark$$

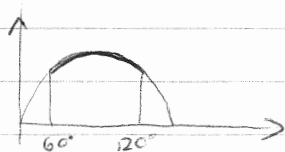
* Far side of moat:

$$\frac{v^2(1+\sqrt{3})\sin 2\theta}{2g} > \frac{v^2(1+\sqrt{3})}{4g} + \frac{v^2}{2g}$$

$$\frac{2v^2(1+\sqrt{3})\sin 2\theta}{4g} > \frac{v^2(3+\sqrt{3})}{4g} \quad \checkmark$$

$$\sin 2\theta > \frac{\sqrt{3}(\sqrt{3}+1)}{2(1+\sqrt{3})}$$

$$\sin 2\theta > \frac{\sqrt{3}}{2} \quad \text{for } 0^\circ < 2\theta < 180^\circ$$



$$60^\circ < 2\theta < 120^\circ$$

$$30^\circ < \theta < 60^\circ \quad \checkmark$$

$$\therefore 0^\circ < \theta < 15^\circ, 30^\circ < \theta < 60^\circ \quad \text{or} \quad 75^\circ < \theta < 90^\circ$$